IoT-Based Environmental Monitoring with Data-Driven Insights

Student Research Proposal • Abidur Rahman Niloy • Data Science • Internet of Things • Cloud Computing

Proposed by	Abidur Rahman Niloy
Project type	Student-led research / prototype study
Estimated duration	10–12 weeks

Project Overview

This project proposes a small IoT monitoring system that collects environmental readings and turns them into simple, useful insights. The student-led study will focus on the complete data journey: sensor collection, storage, visualization, anomaly detection, and basic prediction.

Student perspective
The project is intentionally scoped so that a small student team can learn, experiment, and produce a useful research artifact without assuming access to a large laboratory or proprietary dataset.

Research Problem

IoT projects often demonstrate sensor readings without exploring what can be learned from the data over time. Environmental measurements can contain noise, missing values, sudden changes, and seasonal patterns. Studying these issues can connect IoT hardware with practical data-science methods.

Research Objectives

- Design a low-cost prototype for collecting selected environmental measurements.
- Store and organize time-series readings in a structured database or cloud service.
- Create a dashboard for observing trends and identifying unusual measurements.
- Apply basic statistical or machine-learning methods for anomaly detection.
- Explore whether historical readings can support simple short-term predictions.

Research Questions

RQ1. How reliable are low-cost sensor readings over time?

RQ2. Which anomaly-detection approach works well for the collected data?

RQ3. How do missing readings and connectivity interruptions affect analysis?

RQ4. Can historical measurements provide useful short-term predictions?

Proposed Methodology

Step 1. Choose a small set of sensors and define the sampling interval.

Step 2. Send readings to a local or cloud-based storage layer through a simple IoT communication method.

Step 3. Clean the time-series data and document missing or noisy readings.

Step 4. Develop visualization and baseline anomaly-detection methods.

Step 5. Evaluate the prototype using recorded data and discuss reliability, connectivity, and scalability.

Data / Materials / Experimental Resources

The exact resources will be finalized with the collaborator after a feasibility discussion. The preferred approach is to start with public or self-generated material, keep the experiment small enough to reproduce locally, and record the source, preprocessing decisions, assumptions, and version of each important component.

Initial resource plan: Microcontroller/sensors, time-series storage, dashboard, and recorded environmental readings.

Evaluation Plan

Evaluation will be based on measurable criteria rather than only a successful demo. Depending on the project, this may include classification metrics, detection quality, latency, error analysis, reproducibility, or qualitative review. Baselines will be established before more complex techniques are introduced so that the contribution of each change can be discussed clearly.

Reproducibility and Documentation

The team will keep a simple experiment log containing dataset versions, preprocessing choices, model or system settings, evaluation results, and important observations. Code will be organized so that the main experiment can be rerun without relying on undocumented manual steps. Charts and tables will be generated from saved results where possible.

Expected Challenges

- Limited time and computing resources may require a smaller experimental scope.
- Dataset quality may affect conclusions more than model complexity.
- A strong-looking metric may hide weaknesses that appear during error analysis.
- The research question may need refinement after an initial feasibility experiment.

Planned Mitigation

The project will begin with a small baseline and a feasibility check. If data or hardware limitations make the original scope unrealistic, the team will narrow the research question rather than artificially inflating results. Each limitation will be reported as part of the final research discussion.

Expected Outcomes

- A working low-cost environmental monitoring prototype.
- A structured time-series dataset and data pipeline.
- A simple visualization dashboard.
- An evaluation of anomaly detection and basic predictive methods.

Potential Research Collaboration

I am interested in collaborating with someone who enjoys IoT, data analysis, or cloud systems. A collaborator with experience in sensor integration or time-series modeling could help strengthen both the hardware and research sides of the project.

Technical Stack

Python, SQL, JavaScript, IoT sensors/microcontroller, Git; optional AWS or another cloud platform.

Scope, Ethics, and Responsible Research

Only environmental measurements will be collected. The project will avoid unnecessary personal data. Cloud services will use appropriate access controls, and the prototype will be evaluated within a student-scale budget.

Proposed Work Plan

Phase	Activities
1. Background study	Literature review, problem definition, feasibility check
2. Setup	Dataset/system preparation and baseline implementation
3. Experiments	Development, comparison, tuning, and documentation
4. Evaluation	Metrics, error analysis, limitations, reproducibility
5. Reporting	Results, discussion, documentation, and presentation

What I Am Looking for in a Collaborator

I am looking for someone who is genuinely interested in the research question and willing to contribute to discussion, implementation, experiment design, data analysis, or writing. I would prefer to divide work according to each person's strengths and keep decisions, code, data, and results documented.

Possible Next Step

If a researcher is interested, I would like to first discuss the research question, available resources, and a realistic scope. After that discussion, the proposal can be refined into a clearer study plan with specific responsibilities and milestones.

Note: This document describes a proposed student project. It does not claim completed research, published results, or guaranteed performance. The scope is open to refinement through collaboration.